



Team TROOS

Software Requirements Specification

Introduction

User Stories

- [US-01: Capture Consultant Personal Details](#)
- [US-02: Capture Consultant Skills and Certifications](#)
- [US-03: Set Consultant Availability Status](#)
- [US-04: Set Consultant Cost-to-Company Rate](#)
- [US-05: Consultant profile view](#)
- [US-06: Create a Project Specification](#)
- [US-07: Provide POPIA Consent](#)
- [US-08: Suspend a user](#)
- [US-09: Activate a user](#)
- [US-10: Register new users \(project and consultant managers\)](#)
- [US-11: Set consultancy-wide configurations](#)
- [US-12: Set project-wide configurations](#)
- [US-13: Match consultants to projects](#)

Functional and Non-Functional Requirements

Functional Requirements

- [R1: Authentication and Authorisation](#)
- [R2: Consultant Profile Management](#)
- [R3: Project Specification Management](#)
- [R4: Fit Score Matching Engine](#)
- [R5: Two-Tier Configurable Scoring Algorithm](#)
- [R6: Placement Recommendation Dashboard](#)
- [R7: AI-Powered CV Parsing](#)

Non-functional requirements

- [NFR1: Configurability](#)
- [NFR2: POPIA Compliance](#)
- [NFR3: Usability](#)
- [NFR4: Performance](#)
- [NFR5: Security](#)
- [NFR6: Availability](#)
- [NFR7: Scalability](#)
- [NFR8: Maintainability](#)
- [NFR9: Auditability](#)
- [NFR10: Data Integrity](#)

High-level Use Cases

- [UC1: Login](#)
- [UC2: Create a consultant profile](#)
- [UC3: Upload and parse a CV](#)
- [UC4: Create a project specification](#)
- [UC5: Configure firm-wide scoring defaults](#)

[UC6: Configure project-level scoring override](#)

[UC7: Run a consultant match](#)

[UC8: View placement recommendations](#)

[UC9: Place a consultant](#)

[Diagrams](#)

[Domain Model](#)

[Use Case Diagrams](#)

Introduction

This Software Requirements Specification (SRS) document details the functional and non-functional requirements for the system being developed for Team TROOS. The primary purpose of the system is to streamline the placement of consultants on client engagements through a centralised and automated platform.

User Stories

US-01: Capture Consultant Personal Details

As a consultant manager, I want to create a consultant profile so that all relevant consultant information is captured in a single workflow.

Acceptance Criteria:

Personal Details

- The form accepts the following: first name, surname, nationality, date of birth and ID number.
- If the consultant is a foreign national, a valid work permit expiration date must be provided
- The system validates that the consultant is older than 15 years
- The system validates that the work permit (if required) has not expired

Contact Details

- The form accepts a phone number and email address
- The phone number must be exactly 10 digits and numeric
- Email must follow a valid format (contain “@” and domain structure)
- Both fields are mandatory

Location Details

- The form accepts a town/city and province
- The province must be selected from a predefined list of South African provinces
- Area code (if included) must be numeric and valid length

Definition of Done:

- All validation rules are covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

US-02: Capture Consultant Skills and Certifications

As a consultant manager, I want to capture a consultant's skills, confidence levels, years of experience, and certifications so that we can accurately match capable consultants to suitable projects.

Acceptance Criteria:

- The consultant manager can add one or more skills to a consultant's profile
- For each skill, the consultant manager must select a confidence level from a predefined set: low, standard and high.
- For each skill, the consultant manager must enter the number of years of experience as a positive whole number
- The consultant manager can upload certified copies of certifications as supporting documents (accepted formats: PDF, JPG, PNG)
- Multiple skills can be added in a single session
- A skill entry cannot be saved without a confidence level and years of experience specified
- Uploaded certification documents are stored and linked to the relevant skill

Definition of Done:

- File upload functionality tested with valid and invalid file types and sizes
- Code has been peer-reviewed and approved
- No open bugs against this story

US-03: Set Consultant Availability Status

As a consultant manager, I want to set a consultant's availability status so that only available consultants are considered during project matching.

Acceptance Criteria:

- A toggle control is displayed on the consultant's profile with two states: Available and Unavailable
- The toggle reflects the consultant's current status when the profile is opened
- Changing the toggle immediately updates the consultant's status in the system
- The updated status is visible to project managers when browsing consultants
- An unavailable consultant does not appear in project matching results.

Definition of Done:

- Toggle state change is covered by automated tests, including edge cases (e.g., rapid toggling)
- Code has been peer-reviewed and approved
- No open bugs against this story

US-04: Set Consultant Cost-to-Company Rate

As a consultant manager, I want to set a consultant's cost-to-company rate so that project budgets can be accurately assessed during the matching process.

Acceptance Criteria:

- The consultant manager can enter an hourly cost-to-company rate for a consultant
- The rate must be a positive numeric value; the form rejects zero or negative values
- The currency is displayed alongside the rate field (Rands is the default).
- The rate is stored against the consultant's profile and visible to project managers during matching
- The consultant manager can update the rate at any time; a history of rate changes is maintained

Definition of Done:

- Rate validation is covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

US-05: Consultant profile view

As a consultant, I want to view my own profile so that I can verify that my information is accurate and request corrections if needed.

Acceptance Criteria:

- The consultant can view their personal details, contact details, location, skills, confidence levels, certifications, and availability status on a single profile page
- The consultant can flag any field as incorrect and submit a correction request with a note explaining the issue
- The consultant cannot directly edit fields that are controlled by consultant manager (e.g. cost-to-company rate, availability status)
- The consultant receives a confirmation message once a correction request has been submitted
- The consultant manager is notified of the correction request for review and action

Definition of Done:

- Correction request workflow tested end-to-end, including consultant manager notification
- Code has been peer-reviewed and approved
- No open bugs against this story

US-06: Create a Project Specification

As a project manager, I want to create a project specification so that the system can match the most suitable consultants to the project.

Acceptance Criteria:

- The project manager can create a new project by entering the following: project name, description, required skills, minimum competency level per skill, location, start date, expected duration, team size, and available client billing budget
- At least one required skill with a minimum competency level must be added before the project can be saved
- A skill must be specified, whether it's mandatory for a project
- Team size must be a positive whole number
- The budget must be a positive numeric value in rands
- Start date cannot be set in the past

Definition of Done:

- All field validations are covered by automated unit tests

- Code has been peer-reviewed and approved
- No open bugs against this story

US-07: Provide POPIA Consent

As a consultant, I want to provide consent for my personal data to be collected and processed so that my profile can be legally stored and used.

Acceptance criteria:

- The consultant must explicitly accept a consent agreement before profile creation
- The consent clearly states:
 - o What data is collected
 - o Why it is collected
 - o How it will be used
- The system prevents profile creation if consent is not given

Definition of done:

- All field validations are covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

US-08: Suspend a user

As an admin, I want to be able to suspend users to restrict access when necessary

Acceptance criteria:

- The admin must be logged in to access the suspend functionality.
- The target user must currently be in an activated state before suspension is allowed.
- When the admin clicks the suspend button, the user's status changes to suspended.
- Once suspended, the suspend button should change to an activate button, allowing reversal.
- A confirmation message should be displayed after suspension.

Definition of done:

- All field validations are covered by automated unit tests
- Code has been peer-reviewed and approved

- No open bugs against this story

US-09: Activate a user

As an admin, I want to be able to activate a suspended users to regain access when necessary

Acceptance criteria:

- The admin must be logged in to access the activate functionality.
- The target user must currently be in an suspended state before activation is allowed.
- When the admin clicks the activate button, the user's status changes to active.
- Once activated, the activate button should change to an suspend button, allowing reversal.
- A confirmation message should be displayed after activation .

Definition of done:

- All field validations are covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

US-10: Register new users (project and consultant managers)

As an admin, I want to register consultant managers and project managers so that consultants and projects are handled properly.

Acceptance criteria:

- The admin must be logged in to access the registration functionality.
- The admin can enter the full name, email, and specify the role (consultant manager or project manager)
- Email must follow a valid format (contain "@" and domain structure)
- The system should prevent duplicate email addresses (no two users with the same email).
- All required files (name, email, role) must be filled in before submission.
- If validation fails the system should display an error message and not create the user.

Definition of done:

- All field validations are covered by automated unit tests

- Code has been peer-reviewed and approved
- No open bugs against this story

US-11: Set consultancy-wide configurations

As an admin, I want to be able to set the consultancy-wide configurations for the scoring engine so that I can set baseline factor weights and inclusions.

Acceptance criteria:

- The admin must set weights for each activated factor.
- Deactivated factors shouldn't be considered into the scoring engine.
- The weights should add up to 100%

Definition of done:

- All field validations are covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

US-12: Set project-wide configurations

As a project manager, I want to be able to set project-level configurations for the scoring engine so that factors align with the needs of the project.

Acceptance criteria:

- The project manager must set weights for each activated factor.
- The deactivated factor shouldn't be counted in the scoring engine.
- The weights should add up to to 100%.

Definition of done:

- All field validations are covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

US-13: Match consultants to projects

As a project manager, I want to be able to match consultants to projects via the scoring engine so that the best consultant suited for the project is picked.

Acceptance criteria:

- The project-wide configuration must be set before running the scoring engine.
- Only the project manager assigned to the target project can run the scoring engine.
- An error message should be displayed if a project manager tries to run the scoring engine on a project they haven't been assigned to .
- A ranked recommendation of consultants should be displayed upon successful matching.

Definition of done:

- All field validations are covered by automated unit tests
- Code has been peer-reviewed and approved
- No open bugs against this story

Functional and Non-Functional Requirements

Functional Requirements

R1: Authentication and Authorisation

The system must control access based on user roles. Four roles are defined: Admin, Project Manager, Consultant Manager and Consultant. Role assignment is managed by the Admin. Each role has a defined set of permissions that cannot be overridden by the user.

R1.1: User Login

R1.1.1 The system shall allow users to log in securely using their email address and password.

R1.1.2 The system shall redirect the user to their role-appropriate dashboard upon successful login.

R1.1.3 The system shall display a clear error message when login credentials are invalid, without specifying whether the email or password is incorrect.

R1.1.4 The system shall invalidate the user's session after 8 hours of inactivity and require re-authentication.

R1.2: User Logout

R1.2.1 The system shall allow any authenticated user to log out of their active session at any time.

R1.2.2 The system shall invalidate the user's session token immediately upon logout.

R1.3: Role-Based Access Control

R1.3.1 The system shall enforce four distinct user roles: Admin, Project Manager, Consultant Manager and Consultant.

R1.3.2 The system shall restrict firm-wide scoring configuration to Admin users only. Project Manager, Consultant Manager and Consultant shall not access or modify firm-wide scoring defaults.

R1.3.3 The system shall restrict project specification creation, editing, match run execution, and project-level scoring configuration to Project Manager users.

R1.3.4 The system shall restrict consultant profile creation, editing, and CV upload to Consultant Manager users and subject to the Manager's approval, the system shall allow the consultant to edit their profile.

R1.3.5 The system shall restrict the visibility of consultant cost-to-company rates to Admin and Consultant Manager users only. Project Managers shall not see cost-to-company values.

R1.3.6 The system shall return a 403 Forbidden response for any request by a user attempting to access a Project or action outside their role permissions.

R2: Consultant Profile Management

The system maintains a centralised pool of consultant profiles. Each profile captures the data required by the scoring engine to evaluate a consultant against a project. Profiles are created and maintained by Consultant Managers. All authenticated users can view profiles, subject to field-level access controls.

R2.1: Profile Creation

R2.1.1 The system shall allow a Consultant Manager to create a consultant profile capturing the following fields: full name, email address, city and province of residence, and cost-to-company rate (stored in ZAR per hour).

R2.1.2 The system shall allow a Consultant Manager to add one or more skills to a consultant profile. Each skill entry shall capture the skill name (selected from the

firm's skills reference list), years of experience as a whole number, and a confidence level on a scale of 1 to 10.

R2.1.3 The system shall automatically derive and store a competency level for each skill entry using a weighted formula applied to years of experience and confidence level. The derived level shall be one of: Beginner, Intermediate, or Expert.

R2.1.4 The system shall display the derived competency level to the Consultant Manager in real time as years of experience and confidence values are entered, before the profile is saved.

R2.1.5 The system shall allow a Consultant Manager to add one or more certifications to a consultant profile, capturing the certification name and issuing body.

R2.1.6 The system shall prevent the same skill from being added more than once to a single consultant profile.

R2.1.7 The system shall save the consultant profile and all associated skills and certifications within a single database transaction. If any part of the save fails, no data shall be persisted.

R2.2: Profile Editing

R2.2.1 The system shall allow a Consultant Manager to edit any field on an existing consultant profile, including personal details, skills, and certifications.

R2.2.2 The system shall recompute and update the competency level for any skill where the years of experience or confidence level is changed during an edit.

R2.2.3 The system shall replace the consultant's full skill set atomically on update. The incoming skill list replaces all existing skill entries in a single transaction.

R2.3: Profile Viewing

R2.3.1 The system shall allow all authenticated users to view a paginated list of all consultant profiles, displaying each consultant's name, location, availability status, and primary skills.

R2.3.2 The system shall allow all authenticated users to view the full detail of an individual consultant profile, including all skills with competency levels, certifications, and availability.

R2.3.3 The system shall display the cost-to-company rate on the profile detail view only to users with Admin or Consultant Manager roles. The field shall not be visible or returned in any API response for Project Manager users.

R2.4: Search and Filtering

R2.4.1 The system shall allow users to search for consultants by name using a text input that performs a case-insensitive partial match.

R2.4.2 The system shall allow users to filter the consultant list by one or more of the following: skill, competency level, city, and availability status. Multiple filters shall be applied simultaneously using AND logic.

R3: Project Specification Management

A project specification defines the requirements a consultant must meet to be placed on a client engagement. Project specifications are the input to the matching engine. They are created and managed by Project Managers.

R3.1: Project Creation

R3.1.1 The system shall allow a Project Manager to create a project specification capturing the following fields: project name, client name, project location (city and province), start date, end date, team size, required consultant allocation as a percentage of full-time capacity, and available client billing budget in ZAR per hour.

R3.1.2 The system shall allow a Project Manager to add one or more skill requirements to a project specification. Each skill requirement shall capture the skill name (selected from the firm's skills reference list), the minimum competency level required (Beginner, Intermediate, or Expert), and a mandatory flag indicating whether possession of this skill is a hard requirement for consideration.

R3.1.3 The system shall prevent the same skill from being added more than once to a single project specification and shall display a clear error if this is attempted.

R3.1.4 The system shall save the project specification and all associated skill requirements within a single database transaction. If any part of the save fails, no data shall be persisted.

R3.1.5 The system shall set the initial status of a newly created project to Open.

R3.2: Project Editing

R3.2.1 The system shall allow a Project Manager to edit any field on an existing project specification.

R3.2.2 The system shall allow a Project Manager to add, edit, or remove individual skill requirements from an existing project specification.

R3.3: Project Viewing

R3.3.1 The system shall allow all authenticated users to view a list of all project specifications, displaying the project name, client name, location, status, start date, and number of required skills.

R3.3.2 The system shall allow all authenticated users to view the full detail of an individual project specification, including all skill requirements, minimum competency levels, and mandatory flags.

R3.4: Project Status Management

R3.4.1 The system shall support the following project statuses: Open, In Progress, and Closed.

R3.4.2 The system shall allow a Project Manager to manually update a project's status through the project edit interface.

R3.4.3 The system shall restrict match run execution to projects with a status of Open. Attempting to run a match against an In Progress or Closed project shall return a clear error to the user.

R4: Fit Score Matching Engine

The matching engine computes a weighted fit score for every consultant in the pool against a selected project. The score is a number between 0 and 100, where a higher score indicates a better fit. The engine incorporates five factors, each contributing a weighted sub-score. The weights and factor inclusions are determined by the active scoring configuration at the time of the match run.

R4.1: Match Execution

R4.1.1 The system shall allow a Project Manager to trigger a match run for any project with a status of 'Open'.

R4.1.2 The system shall compute a fit score for every consultant in the pool against the selected project and return a ranked list ordered by fit score, highest first.

R4.1.3 The system shall incorporate the following five factors in the fit score computation: skill alignment, competency level match, availability, cost fit, and geographic fit.

R4.1.4 The system shall apply the project-level scoring configuration for match runs where a project-level override exists and the firm-wide default configuration otherwise.

R4.1.5 The system shall store each match run result with a timestamp and the scoring configuration snapshot used to support auditability of placement recommendations.

R4.2: Skill Alignment

R4.2.1 The system shall compute the skill alignment sub-score as the proportion of the project's required skills possessed by the consultant, expressed as a percentage.

R4.2.2 The system shall exclude any consultant from the ranked results who does not possess at least one skill flagged as mandatory on the project specification, unless hard exclusion is disabled in the active scoring configuration.

R4.3: Competency Match

R4.3.1 The system shall compute the competency match sub-score by evaluating the consultant's competency level against the minimum required level for each required skill.

R4.3.2 The system shall award full marks on the competency sub-score for any skill where the consultant's competency level meets or exceeds the minimum required level.

R4.3.3 The system shall award partial marks on the competency sub-score for any required skill where the consultant's competency level is below the minimum required level. Partial marks are calculated proportionally based on the distance from the required level.

R4.3.4 The system shall award zero marks on the competency sub-score for any required skill that the consultant does not possess.

R4.4: Availability

R4.4.1 The system shall compute the availability sub-score based on the consultant's remaining capacity during the project period, derived from their active placement records.

R4.4.2 The system shall award full marks on the availability sub-score where the consultant's available capacity meets or exceeds the project's required allocation percentage.

R4.4.3 The system shall award partial marks on the availability sub-score where the consultant's available capacity is greater than zero but below the required allocation percentage.

R4.4.4 The system shall award zero marks on the availability sub-score where the consultant has no remaining capacity during the project period.

R4.5: Cost Fit

R4.5.1 The system shall compute the cost fit sub-score by comparing the consultant's cost-to-company rate against the project's available billing budget per hour.

R4.5.2 The system shall award full marks on the cost fit sub-score where the consultant's cost-to-company rate is less than or equal to the project's billing budget.

R4.5.3 The system shall award reduced marks on the cost fit sub-score where the consultant's cost-to-company rate exceeds the project's billing budget.

R4.6: Geographic Fit

R4.6.1 The system shall compute the geographic fit sub-score based on the proximity of the consultant's city of residence to the project location, using a region-based proximity model.

R4.6.2 The system shall award full marks on the geographic fit sub-score where the consultant and project are in the same city.

R4.6.3 The system shall award partial marks on the geographic fit sub-score where the consultant and project are in the same province but different cities.

R4.6.4 The system shall award reduced marks on the geographic fit sub-score where the consultant and project are in different provinces.

R5: Two-Tier Configurable Scoring Algorithm

The scoring engine must not contain any hardcoded business logic. All factor weights, factor inclusions, and hard exclusion settings must be stored in the database and configurable through the admin interface. The configuration operates at two levels: a firm-wide default and a project-level override. Project-level settings always take precedence over firm-wide defaults for that match run.

R5.1: Firm-Wide Scoring Configuration

R5.1.1 The system shall allow an Admin to configure firm-wide default scoring factor weights through a dedicated configuration interface.

R5.1.2 The system shall allow an Admin to activate or deactivate individual scoring factors at the firm-wide level. Deactivated factors are excluded from all match run computations using the firm-wide default.

R5.1.3 The system shall validate that the sum of all active scoring factor weights equals exactly 100% and shall reject the configuration with a clear error message if validation fails.

R5.1.4 The system shall seed the firm-wide scoring configuration on initial setup with the following default weights (subject to change): Skill Alignment 40%, Competency Match 30%, Availability 15%, Cost Fit 10%, and Geographic Fit 5%, with all five factors active and hard exclusion enabled.

R5.1.5 The system shall record a timestamped audit log entry every time the firm-wide scoring configuration is changed, capturing the previous values, the new values, and the identity of the Admin who made the change.

R5.2: Project-Level Scoring Override

R5.2.1 The system shall allow a Project Manager to configure a project-level scoring override for a specific project from the project detail view.

R5.2.2 The system shall allow a Project Manager to activate, deactivate, or reweight individual scoring factors at the project level. Only factors being explicitly changed need to be stored; all other factors inherit their values from the firm-wide default.

R5.2.3 The system shall apply the project-level scoring configuration in place of the firm-wide default for all match runs executed against that project.

R5.2.4 The system shall fall back to the firm-wide default value for any scoring factor not explicitly overridden at the project level.

R5.2.5 The system shall validate that the sum of all active factor weights in a project-level override equals 100% before saving.

R6: Placement Recommendation Dashboard

The dashboard is the primary output of the system. It presents the results of a match run in a format that is explainable and auditable – a manager must be able to understand exactly why a consultant ranked where they did without technical knowledge of the scoring algorithm.

R6.1: Results Display

R6.1.1 The system shall display match run results as a ranked list of consultants ordered by total fit score, from highest to lowest.

R6.1.2 The system shall display the total fit score for each consultant as a number between 0 and 100.

R6.1.3 The system shall display a per-factor score breakdown for each consultant in the ranked list, showing the sub-score and the weight applied for each active scoring factor.

R6.1.4 The system shall display each consultant's name, city, current availability, and their competency level for each matched skill alongside their score.

R6.1.5 The system shall display the scoring configuration used for the match run, including factor weights, so that the basis for the ranking is transparent.

R6.2: Explainability

R6.2.1 The system shall provide a plain-language explanation for each consultant's ranking, describing the primary reasons they ranked where they did relative to the project requirements.

R6.2.2 The system shall clearly indicate for each consultant which mandatory skills they possess and which they are missing.

R6.2.3 The system shall indicate what changes to a consultant's profile or the project configuration would be required for a lower-ranked consultant to improve their position.

R6.3: Placement Action

R6.3.1 The system shall allow a Project Manager to mark a consultant as placed on a project directly from the recommendation dashboard.

R6.3.2 The system shall create a placement record when a consultant is marked as placed, capturing the consultant identifier, project identifier, start date, planned end date, and allocation percentage.

R6.3.3 The system shall immediately reduce the placed consultant's available capacity to reflect the new placement, affecting all subsequent match runs for overlapping project periods.

R7: AI-Powered CV Parsing

CV parsing allows a Consultant Manager to upload a consultant's CV and have the system automatically extract structured data from it using an LLM. The extracted data is always presented for human review before being applied to the profile. No LLM output is ever written directly to a profile without explicit manager confirmation.

R7.1: File Upload

R7.1.1 The system shall allow a Consultant Manager to upload a consultant's CV in PDF format from the consultant profile creation or edit screen.

R7.1.2 The system shall store the uploaded CV as binary data in the database, associated with the consultant's record, retaining the original filename and MIME type.

R7.1.3 The system shall reject uploads that exceed 10MB or are not in PDF format, displaying a clear error message to the user.

R7.2: Extraction and Review

R7.2.1 The system shall submit the uploaded CV to an LLM-based extraction service to identify skills, years of experience per skill, competency signals, certifications, and total years of professional experience.

R7.2.2 The system shall present all extracted data to the Consultant Manager in a structured review screen before any changes are applied to the consultant's profile.

R7.2.3 The system shall display the LLM's reasoning for each inferred competency level alongside the extracted skill data in the review screen, allowing the manager to understand and evaluate the inference.

R7.2.4 The system shall allow the Consultant Manager to accept, edit, or remove any individual extracted item during the review step.

R7.2.5 The system shall apply the reviewed and confirmed extraction data to the consultant's profile only after the Consultant Manager explicitly confirms the submission. Closing or cancelling the review screen shall discard all extracted data without modifying the profile.

R7.2.6 The system shall flag any skill extracted from the CV that does not match an existing entry in the firm's skills reference list. Flagged skills shall be presented separately in the review screen and shall require an Admin to approve their addition to the reference list before they can be saved to the consultant profile.

R7.2.7 The system shall apply the competency derivation formula to any skill accepted from the extraction, computing the competency level from the extracted years of experience and confidence signal before saving to the profile.

REQUIREMENTS TRACEABILITY MATRIX

Each row maps a functional requirement to the use cases that exercise it. A requirement not mapped to any use case indicates a gap in use case coverage and should be investigated.

FR	Description	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	UC9
R1.1.1	Login with email and password	x								
R1.1.2	Redirect to dashboard on login	x								
R1.2.1	Logout of active session									
R1.3.1	Four roles: Admin, Project Manager, Consultant Manager, Consultant	x								
R1.3.2	Restrict scoring config to Admin					x				
R1.3.3	Restrict project creation to Project Manager				x		x			
R1.3.4	Restrict profile and CV upload to Consultant Manager		x	x						
R1.3.5	Restrict cost-to-company visibility		x							
R2.1.1	Create consultant profile with personal details		x							

R2.1.2	Add skills with years and confidence level		x							
R2.1.3	Derive competency level from years and confidence		x	x						
R2.1.4	Add certifications to profile		x	x						
R2.2.1	Edit any field on an existing profile									
R2.2.2	Atomic skill replacement in single transaction									
R2.3.1	View list of all consultant profiles						x	x		
R2.3.2	View full details of individual profile							x		
R2.4.1	Search consultants by name						x			
R2.4.2	Filter by skill, competency, location, availability						x			
R3.1.1	Create project with all specification fields				x					
R3.1.2	Add skill requirements with competency and mandatory				x					
R3.1.3	Prevent duplicate skills on same project				x					
R3.1.4	Atomic save of project and skill requirements				x					
R3.3.1	Display project status on list and detail views									
R3.4.1	Support Open, In Progress, and Closed statuses									
R3.4.3	Restrict match runs to Open projects only						x			
R4.1.1	Trigger match run for an Open project						x			
R4.1.2	Compute fit score for every consultant in pool						x	x		
R4.2.1	Compute skill alignment sub-score						x	x		
R4.2.2	Exclude consultants missing mandatory skills						x	x		
R4.3.1	Compute competency match sub-score						x	x		
R4.4.1	Compute availability from placement records						x	x		
R4.5.1	Compute cost fit sub-score						x	x		
R4.6.1	Compute geographic fit sub-score						x	x		
R5.1.1	Configure firm-wide default scoring weights					x				

R5.1.3	Validate weights sum to 100% before saving					x				
R5.1.5	Audit log on scoring config change					x	x			
R5.2.1	Configure project-level scoring overrides						x			
R5.2.3	Apply project-level config for that project's runs						x	x		
R6.1.1	Display ranked list of consultants by fit score								x	
R6.1.3	Display per-factor score breakdown								x	
R6.2.1	Plain-language explanation per consultant								x	
R6.2.2	Indicate missing mandatory skills								x	
R6.3.1	Mark consultant as placed from dashboard									x
R6.3.2	Create placement record on placement action									x
R7.1.1	Upload CV in PDF format			x						
R7.2.1	Submit CV to LLM extraction service			x						
R7.2.2	Present extracted data for manager review			x						
R7.2.5	Apply extraction only after confirmed submission			x						
R7.2.6	Flag unrecognised skills for admin review			x						

Non-functional requirements

NFR1: Configurability

No consultancy-specific business logic may be hardcoded in the application. All placement rules, scoring factors, factor weights, hard exclusion settings, and competency derivation Consultant thresholds must be configurable through the admin interface and stored in the database. A developer should never need to deploy a code change to modify how the scoring engine evaluates consultants. Violations of this requirement shall be treated as critical defects and shall be rejected at code review.

NFR2: POPIA Compliance

All personal consultant data - including full name, email address, location, cost-to-company rate, skills, certifications, and CV files - must be handled in compliance with the Protection of Personal Information Act (POPIA). Specifically:

Data storage: all personal data must be stored in a hosting environment within the Republic of South Africa or in a jurisdiction approved under POPIA.

Access controls: cost-to-company rate and CV files must only be accessible to users with Admin or Consultant Manager roles, enforced at both the API and database levels.

Consent: the system must document the basis on which consultant personal data is collected and processed.

Retention: a data retention policy must be defined and documented, specifying how long consultant data is retained and the process for data deletion upon request.

NFR3: Usability

The user interface must be operable by users with no technical background. Specifically: the scoring configuration interface must express factor weights as percentages with slider controls rather than raw numeric inputs; the placement recommendation dashboard must communicate scores in plain language alongside numeric values; all form validation errors must be expressed in plain language that describes what the user needs to do to resolve the issue, not what failed internally.

NFR4: Performance

The system shall return match run results for a consultant pool of up to 200 profiles within 5 seconds under normal operating conditions. All other standard user interactions – loading a consultant list, saving a profile, viewing a project – shall be completed within 3 seconds under normal network conditions. These consultant hold times apply to the server-side processing time, excluding network latency on the client side.

NFR5: Security

All data transmitted between the client and server shall be encrypted using TLS 1.2 or higher. Sensitive fields at rest – including cost-to-company rate and CV file content – shall be encrypted in the database. Session tokens shall be stored in HTTP-only

cookies to prevent client-side script access. All API endpoints shall validate the authenticated user's role before processing any request, server-side, regardless of client-side access controls.

NFR6: Availability

The system shall maintain an uptime of at least 99% during business hours (07h00 to 20h00 SAST, Monday to Friday). Planned maintenance windows must be scheduled outside these hours and communicated to users in advance.

NFR7: Scalability

The system architecture shall support a minimum of 500 consultant profiles and 100 concurrent open projects without response times exceeding the Consultant roles are defined in NFR4. The data model shall include tenant isolation fields from day one to support multi-tenancy in a future phase without requiring a schema migration.

NFR8: Maintainability

All architectural decisions shall be documented in Architecture Decision Records, each capturing the decision made, the context in which it was made, the alternatives considered, and the tradeoffs accepted. The codebase shall conform to the agreed coding standards document. Every module boundary shall be enforced: no module may import a repository from another module. These rules shall be verified at code review.

NFR9: Auditability

The system shall maintain a tamper-evident audit log of the following events: firm-wide scoring configuration changes (capturing before and after values and the acting user), placement actions (capturing the consultant, project, acting user, and timestamp), and CV extraction events (capturing which CV was processed, what was extracted, and what the manager confirmed or modified). Audit log entries shall never be deleted.

NFR10: Data Integrity

All write operations that involve more than one table - including consultant profile creation with skills, project creation with skill requirements, and CV extraction confirmation - shall be executed within a database transaction. If any step in the

operation fails, the entire transaction shall be rolled back. No partial writes shall be persisted under any circumstances.

High-level Use Cases

TUCBW = The use case begins when.

TUCEW = The use case ends when.

UC1: Login

TUCBW: The user navigates to the ConsultIQ login page and enters their email address and password, then clicks Log In.

TUCEW: The user is authenticated and redirected to their role-appropriate dashboard. An active session is established.

UC2: Create a consultant profile

TUCBW, a consultant manager, clicks 'Create Profile' on the consultant list page and begins entering consultant details.

TUCEW, the consultant manager, has entered all required fields, added at least one skill with years of experience and a confidence level, reviewed the derived competency levels, and clicked Save. The consultant profile is visible in the consultant list.

UC3: Upload and parse a CV

TUCBW, a consultant manager, clicks 'Upload CV' on a consultant's profile page and selects a PDF file from their device.

TUCEW, the consultant manager, has reviewed all extracted skills, competency inferences, and certifications; made any necessary corrections; and confirmed the submission, and the consultant's profile has been updated to reflect the confirmed data.

UC4: Create a project specification

TUCBW, a project manager, clicks 'Create Project' on the project list page and begins entering project details.

TUCEW, the project manager, has entered all required project fields, added at least one skill requirement with a minimum competency level and mandatory flag, and clicked Save. The project is visible in the project list with a status of Open.

UC5: Configure firm-wide scoring defaults

TUCBW, an admin navigates to the Scoring Configuration page and views the current firm-wide default factor weights.

TUCEW, the admin has adjusted one or more factor weights or toggled a factor's active state; confirmed that the weights sum to 100%; and saved the configuration. An audit log entry has been created capturing the change.

UC6: Configure project-level scoring override

TUCBW, a project manager, opens a project specification and clicks 'Configure Scoring Override'.

TUCEW, the project manager, has set one or more project-level factor weight overrides, confirmed that the active weights sum to 100%, and saved the override. Subsequent match runs for this project will use the project-level configuration.

UC7: Run a consultant match

TUCBW, a project manager opens an open project specification and clicks Run Match.

TUCEW, the system has computed a weighted fit score for every consultant in the pool against the selected project and returned a ranked list of results on the placement recommendation dashboard.

UC8: View placement recommendations

TUCBW, a consultant manager, views the ranked results on the placement recommendation dashboard following a match run.

TUCEW, the consultant manager, has reviewed the ranked list, examined the per-factor score breakdown and plain-language explanation for one or more consultants, and understands the basis for the ranking.

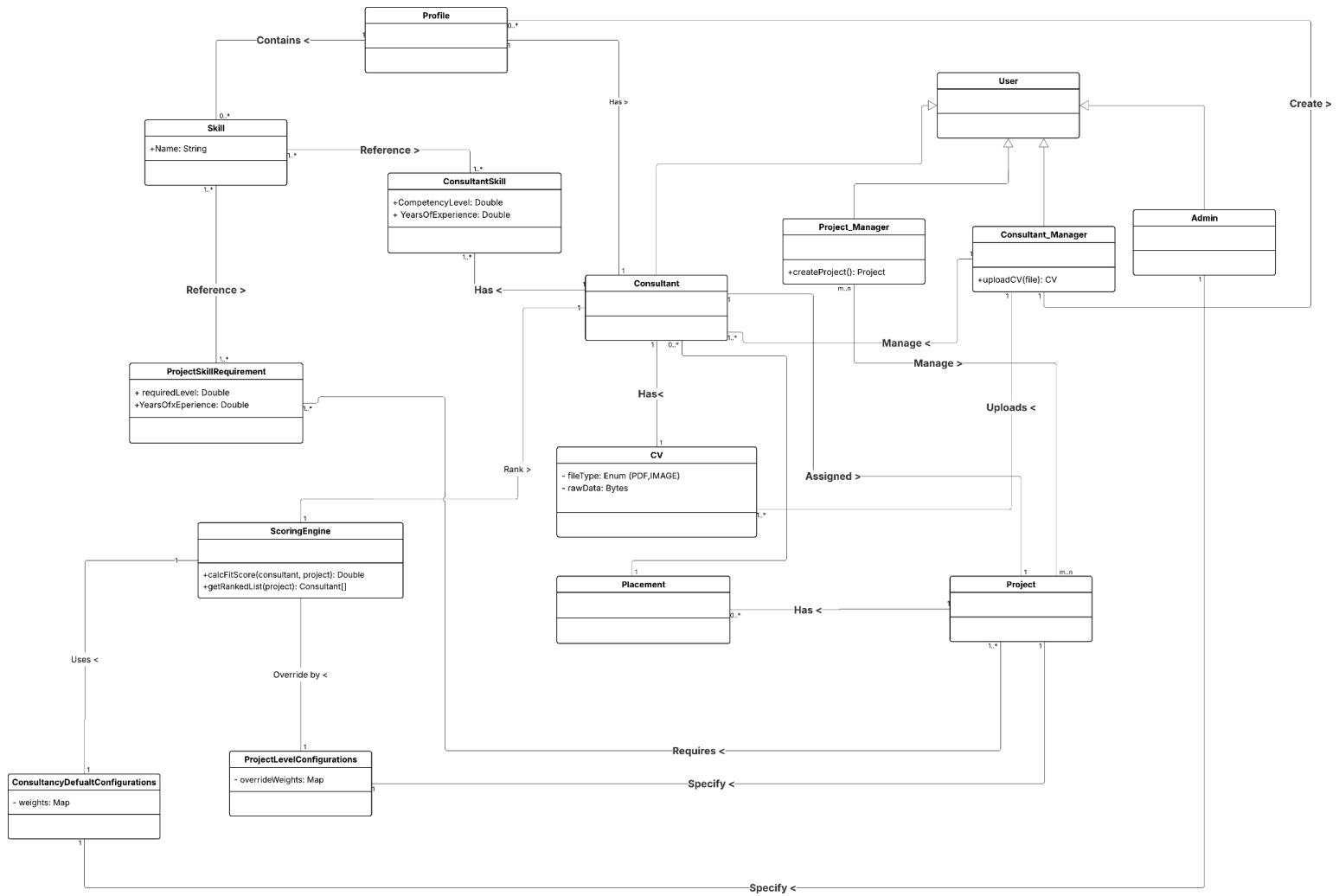
UC9: Place a consultant

TUCBW, a consultant manager selects a consultant from the ranked results on the placement recommendation dashboard and clicks 'Place'.

TUCEW, a placement record has been created capturing the consultant, project, start date, planned end date, and allocation percentage. The consultant's available capacity has been updated immediately to reflect the new commitment.

Diagrams

Domain Model



Use Case Diagrams

